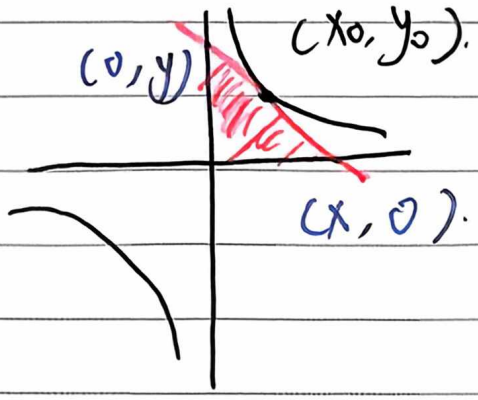
Derivative

$$f'(x) = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f = \frac{d}{dx}y$$

$f(x) = \frac{1}{x}$ calculate the area

$\Downarrow$



$k = f'(x) = -\frac{1}{x^2}$

$y - y_0 = k(x - x_0).$

$k = -\frac{1}{x_0^2}$ $\quad y - y_0 = -\frac{1}{x_0^2}(x - x_0)$

① $y = 0$ $\quad 0 - y_0 = -\frac{1}{x_0^2}(x - x_0)$

$\cancel{-\frac{1}{x_0^2}} = -\frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0)$

$x_0 = x - x_0$

$x = 2x_0$

② $x = 0$ $\quad y - \frac{1}{x_0} = -\frac{-x_0}{x_0^2}$

$y = \frac{2}{x_0}$

area $= \frac{1}{2}xy = \frac{1}{2} \cdot 2x_0 \cdot \frac{2}{x_0} = 2.$

$f(x) = x^n \quad f'(x) = n \cdot x^{n-1}$

$(x+\Delta x)^n = (x+\Delta x) \cdots (x+\Delta x)$

$= x^n + nx^{n-1}\Delta x + \text{junk}$

$O((\Delta x)^2)$.

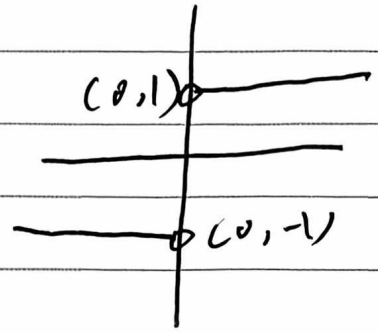
$$f'(x) = \frac{\Delta f}{\Delta x} = \frac{(x+\Delta x)^n - x^n}{\Delta x} = \frac{x^n + nx^{n-1}\Delta x - x^n}{\Delta x}$$

$$= n \cdot x^{n-1}$$

type of Discontinuties.

① Jump Discontinuity.

$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$



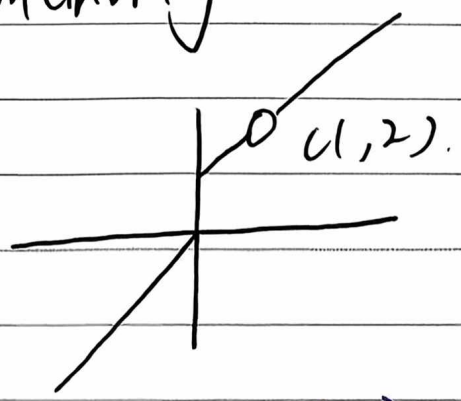
~~$\lim_{x \to 0}$~~ $\lim_{x \to 0^-} f(x) = -1$

$\lim_{x \to 0^+} f(x) = 1.$

$\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x).$

② Removable Discontinuity

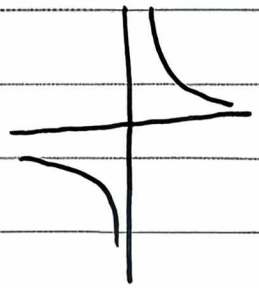
$$f(x) = \frac{x^2-1}{x-1}$$



$x=1$ is Discontinuous. $f(1)$ is not defined

③ Infinite Discontinuity.

$f(x) = \frac{1}{x}$



$\lim_{x \to 0^-} f(x) = -\infty$ $x = 0$

$\lim_{x \to 0^+} f(x) = +\infty$ The ~~limist~~ limit is not exist

④ Ugly Discontinuity.

$f(x) = \sin(\frac{1}{x})$.

$\lim_{x \to 0^+} \frac{1}{x} = +\infty$

$\sin(+\infty)$ is always changing.

$\lim_{x \to 0} \sin(\frac{1}{x})$ is not exist

Overall!

$f(x)$ is continuous at $x=a$ if and only if

① $f(a)$ is defined

② The limit exists $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

③ The limit equals the function value

$\lim_{x \to a} f(x) = f(a)$.

$\cancel{f(x)}$ $f(x) = \sin x$ $\qquad \sin(a+b) =$

$f'(x) = \cos x$ $\qquad \sin a \cos b + \cos a \sin b$

$f(x) = \cos x$ $\qquad \cos(a+b) =$

$f'(x) = -\sin x$ $\qquad \cos a \cos b - \sin a \sin b.$

① $f(x) = \sin x$

$$f'(x) = \frac{\sin(x+\Delta x) - \sin x}{\Delta x}$$

$$= \frac{\sin x \cos \Delta x + \cos x \sin \Delta x - \sin x}{\Delta x}$$

$A = 0$ $\qquad B = 1$

$$= \sin x \left(\frac{\cos \Delta x - 1}{\Delta x}\right) + \cos x \left(\frac{\sin \Delta x}{\Delta x}\right)$$

$$= \cos x$$

$\Delta x \to 0$

② $f(x) = \cos x$

$$f'(x) = \frac{\cos(x+\Delta x) - \cos x}{\Delta x}$$

$$= \frac{\cos x \cos \Delta x - \sin x \sin \Delta x - \cos x}{\Delta x}$$

$A = 0$ $\qquad B = 1$

$$= \cos x \left(\frac{\cos \Delta x - 1}{\Delta x}\right) - \sin x \left(\frac{\sin \Delta x}{\Delta x}\right)$$

$$= -\sin x$$

$\Delta x \to 0$

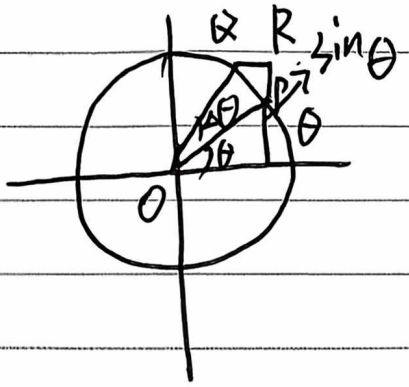
Ⓐ $\lim_{\Delta x \to 0} \frac{\cos\Delta x - 1}{\Delta x}$

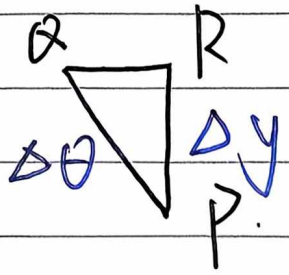
$$= \frac{\cos\Delta x - 1}{\Delta x} \cdot \frac{\cos\Delta x + 1}{\cos\Delta x + 1}$$

$$= \frac{-\sin^2\Delta x}{\Delta x(\cos\Delta x + 1)} = -\frac{\sin\Delta x}{\Delta x} \cdot \frac{\sin\Delta x}{\cos\Delta x + 1}$$

$$\xrightarrow{\Delta x \to 0} = -1 \cdot \frac{0}{1+1} = 0.$$

Ⓑ $\lim_{\Delta x \to} \frac{\sin\Delta x}{\Delta x} = 1$





$OP \perp PQ$

$PQ \approx \Delta\theta$

$\angle QPR \approx \theta$

$PR = \Delta y$

$$\frac{PR}{PQ} = \frac{\Delta y}{\Delta\theta} = \cos\theta$$

$(uv)'$

Product rule $(\cancel{u+v})' = u'v + uv'$

proof: $\Delta(\cancel{u+v}\ uv) = u(x+\Delta x)v(x+\Delta x) - u(x)v(x)$

$= u(x+\Delta x)v(x+\Delta x) + u(x)v(x+\Delta x) - u(x)v(x+\Delta x) - u(x)v(x)$

$= (u(x+\Delta x) - u(x))v(x+\Delta x) + u(x)(v(x+\Delta x) - v(x))$

$= \Delta u\, v(x+\Delta x) + u(x)\Delta v$

$$\frac{\Delta(u+v)}{\Delta x} = \frac{\Delta u\, v(x+\Delta x)}{\Delta x} + \frac{u(x)\Delta v}{\Delta x}$$

$$= \frac{\Delta u}{\Delta x}\cdot v(x+\Delta x) + \frac{\Delta v}{\Delta x}\cdot u(x).$$

$$\Delta x \to 0 = \frac{\Delta u}{\Delta x}\cdot v(x) + \frac{\Delta v}{\Delta x}\cdot u(x)$$

$$= u'v + v'u$$

Quotient rule $\left(\frac{u}{v}\right)' = \frac{u'v - vv'}{v^2}$

proof: $\Delta\left(\frac{u}{v}\right) = \frac{u+\Delta u}{v+\Delta v} - \frac{u}{v} = \frac{uv + \Delta uv - uv - u\Delta v}{v(v+\Delta v)}$

$= \frac{\Delta uv - \Delta vu}{v(v+\Delta v)}$ $\qquad \frac{\Delta\left(\frac{u}{v}\right)}{\Delta x} = \frac{\frac{\Delta u}{\Delta x}\cdot v - \frac{\Delta v}{\Delta x}\cdot u}{v\cdot(v+\Delta v)}$

$$= \frac{u'v - uv'}{v^2}$$

chain rule

$y = f(u)$

$u = g(x)$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$